Def: Let $\alpha, \beta \in \mathbb{Z}^\ell$ be $GL_\ell$ weights and $\sigma \in S_\ell$. The LLT series $\mathcal{L}^\sigma_{\beta/\alpha}(x_1, \dots, x_\ell; q)$ is the infinite formal sum of irreducible $GL_\ell$ characters in which the coeff of $\chi_\lambda$ is defined by

$$\langle \chi_\lambda \rangle \mathcal{L}^{\sigma^{-1}}_{\beta/\alpha}(x; q^{-1}) = \langle E^\sigma_\beta \rangle \chi_\lambda \cdot E^\sigma_\alpha$$

* Since coeff in $E^\sigma_\lambda(x;q)$ are in $q^{-1}$, coeff. in $\mathcal{L}^\sigma_{\beta/\alpha}(x;q)$ are thus polynomials in $q$.

The following theorem makes computation of $\mathcal{L}^\sigma_{\beta/\alpha}$ easier.

Prop: $\mathcal{L}^\sigma_{\beta/\alpha}(x;q) = H_q\left(w_0\left(F^{\sigma^{-1}}_\beta(x;q)\,\overline{E^{\sigma^{-1}}_\alpha(x;q)}\right)\right)$ where $w_0(i) = \ell+1-i \ \forall i \in [\ell]$ (i.e. $w_0$: longest permutation in $S_\ell$).

Proof: $\because$ $\{E^{\sigma^{-1}}_\lambda\}_{\lambda \in \mathbb{Z}^\ell}$ and $\{F^{\sigma^{-1}}_\lambda\}_{\lambda \in \mathbb{Z}^\ell}$ are dual bases

$\therefore$ coeff of $E^{\sigma^{-1}}_\beta(x;q^{-1})$ in $\chi_\lambda \cdot E^{\sigma^{-1}}_\alpha(x;q^{-1}) = \langle F^{\sigma^{-1}}_\beta(x^{-1};q), \chi_\lambda E^{\sigma^{-1}}_\alpha(x;q^{-1}) \rangle_{q^{-1}}$

$$= \langle x^0 \rangle \chi_\lambda F^{\sigma^{-1}}_\beta(x^{-1};q)\, E^{\sigma^{-1}}_\alpha(x;q^{-1}) \prod_{i<j} \frac{1-\frac{x_i}{x_j}}{1-q\frac{x_i}{x_j}}$$

$$w_0\left(\prod_{i<j} \frac{1-\frac{x_j}{x_i}}{1-q\frac{x_j}{x_i}}\right) = \prod_{i>j} \frac{1-\frac{x_j}{x_i}}{1-q\frac{x_j}{x_i}} = \prod_{i<j} \frac{1-\frac{x_i}{x_j}}{1-q\frac{x_i}{x_j}}$$

Then take $w_0$ ($x_i \mapsto x_i^{-1}$) (constant term is not affected by these operations)

$$= \langle x^0 \rangle \overline{\chi_\lambda}\, w_0\left(F^{\sigma^{-1}}_\beta(x;q)\, E^{\sigma^{-1}}_\alpha(x^{-1};q^{-1})\right) \prod_{i<j} \frac{1-\frac{x_i}{x_j}}{1-q\frac{x_i}{x_j}}$$

$w_0(\overline{\chi_\lambda}) = \overline{\chi_\lambda}$ b/c $\chi_\lambda$ is symmetric

Recall §2 (c.f. Extended Delta notes §3.4 P.2 on Lemma 2.3.1)

we have

$\langle \chi_\lambda \rangle \sigma(\Phi(x)) = \langle x^0 \rangle \overline{\chi_\lambda}\, \Phi(x) \prod_{i<j} (1 - \frac{x_i}{x_j})$

$$= \langle \chi_\lambda \rangle \sigma\left(w_0\left(F^{\sigma^{-1}}_\beta(x;q)\, E^{\sigma^{-1}}_\alpha(x^{-1};q^{-1}) \prod_{i<j} \frac{1}{1-q\frac{x_i}{x_j}}\right)\right) = \langle \chi_\lambda \rangle H_q\left(w_0\left(F^{\sigma^{-1}}_\beta(x;q)\, \overline{E^{\sigma^{-1}}_\alpha(x;q)}\right)\right). \quad \square$$

Take the expression and expand as a formal Laurent series in $q$, i.e. expand $\prod \frac{1}{1-q\frac{x_i}{x_j}} = \prod (1 + q\frac{x_i}{x_j} + q^2 \frac{x_i^2}{x_j^2} + \cdots)$

as a Laurent series and apply the Lemma in §2 term by term,

and recombine to get the expression $\sigma\left(w_0\left(F^{\sigma^{-1}}_\beta(x;q)\, E^{\sigma^{-1}}_\alpha(x^{-1};q^{-1}) \prod_{i<j} \frac{1}{1-q\frac{x_i}{x_j}}\right)\right)$

because $\sigma$ is linear

e.g. $\mathcal{L}^{S_1}_{32/11}(x;q) = H_q\left(w_0\left(F^{S_1}_{32}(x;q)\, \overline{E^{S_1}_{11}(x;q)}\right)\right)$ $\quad w_0 = S_1 \in S_2$ as $\ell = 2$; $\quad E^{S_1}_{11} = x^{11} = x_1 x_2$

$$= H_q\left(w_0\left(F^{S_1}_{32}(x_1, x_2; q)\, x_1^{-1} x_2^{-1}\right)\right)$$

$$\overline{E^{S_1 w_0}_{(-3,-2)}(x;q)} = \overline{E^{id}_{(-3,-2)}(x_1, x_2; q)} = \overline{q^{-1} T_1(x^{-2,-3})} = \overline{q^{-1}(q-1)x_1^{-2}x_2^{-3} + q^{-1} \cdot q\, x_1^{-3}x_2^{-2}} = (1-q)x_1^2x_2^3 + x_1^3x_2^2$$

$$= H_q\left(w_0\left((1-q)\, x_1 x_2^2 + x_1^2 x_2\right)\right)$$

$$= H_q\left((1-q)\, x_1^2 x_2 + x_1 x_2^2\right)$$

$$= \sum_{w \in S_2} \left( \frac{(1-q)x_1^2x_2 + x_1x_2^2}{(1-\frac{x_2}{x_1})(1-q\frac{x_1}{x_2})} \right)$$

$$= \frac{(1-q)x_1^2x_2 + x_1x_2^2}{(1-\frac{x_2}{x_1})(1-q\frac{x_1}{x_2})} + \frac{(1-q)x_1x_2^2 + x_1^2x_2}{(1-\frac{x_1}{x_2})(1-q\frac{x_2}{x_1})}$$

$$= \frac{1}{x_1 - x_2}\left[\left((1-q)\, x_1^3x_2 + x_1^2x_2^2\right)\left(1 + q\frac{x_1}{x_2} + q^2\frac{x_1^2}{x_2^2} + \cdots\right) - \left((1-q)\, x_1x_2^3 + x_1^2x_2^2\right)\left(1 + q\frac{x_2}{x_1} + q^2\frac{x_2^2}{x_1^2} + \cdots\right)\right]$$

$$= \frac{1}{x_1 - x_2}\left[(1-q)\left(\left(x_1^3x_2 - x_1x_2^3\right) + q\left(x_1^4 - x_2^4\right) + q^2\left(\frac{x_1^5}{x_2} - \frac{x_2^5}{x_1}\right) + q^3\left(\frac{x_1^6}{x_2^2} - \frac{x_2^6}{x_1^2}\right) + \cdots\right) + x_1^2x_2^2\left(q\,\frac{x_1^2 - x_2^2}{x_1x_2} + q^2\,\frac{x_1^4 - x_2^4}{x_1^2x_2^2} + \cdots\right)\right]$$

$$= (1-q)\left[\overset{\chi_{21}}{x_1x_2(x_1+x_2)} + q\overset{\chi_{30}}{(x_1^3 + x_1^2x_2 + x_1x_2^2 + x_2^3)} + q^2\overset{\chi_{4-1}}{\left(\frac{x_1^5 + x_1^4x_2 + \cdots + x_2^5}{x_1x_2}\right)} + q^3\overset{\chi_{5-2}}{\left(\frac{x_1^7 + x_1^6x_2 + \cdots + x_2^8}{x_1^2x_2^2}\right)} + \cdots\right]$$

$$+ q\left[x_1x_2(x_1+x_2) + q(x_1^3 + x_1^2x_2 + x_1x_2^2 + x_2^3) + q^2\left(\frac{x_1^5 + x_1^4x_2 + \cdots + x_2^5}{x_1x_2}\right) + q^3\left(\frac{x_1^7 + x_1^6x_2 + \cdots + x_2^8}{x_1^2x_2^2}\right) + \cdots\right]$$

$$= \chi_{21} + q\chi_{30} + q^2\chi_{4-1} + q^3\chi_{5-2} + \cdots$$